PART I: FOUNDATIONS

Topic 1: What Is a Time Series?

The Core Concept

A time series is a sequence of data points recorded at **successive, equally-spaced intervals**. In public safety contexts, it is not just about *how many* incidents occurred, but **when** they occurred in relation to each other.

Why It Matters

Imagine a Police Chief asks: "How are we doing with use-of-force?"

- **The single-number answer:** "We had 185 incidents this year." This tells you the volume, but not the story.
- **The time series answer:** "Incidents peaked in July and dropped by December." This reveals a narrative.

The Example

Note: All data in this tutorial series are synthetic, created for teaching purposes only. They do not represent real-world incidents or any specific jurisdiction.

Precinct North, 2023 — Use-of-Force Incidents:

Month	Precinct North (UoF)	Precinct South (UoF)	North Arrests
Jan	12	25	400
Feb	10	22	380
Mar	14	28	450
Apr	15	24	460
May	18	26	550
Jun	22	24	600
Jul	25	27	700
Aug	**(Missing)**	23	680
Sep	19	22	500
Oct	16	**85 ★ Outlier**	420
Nov	14	24	390
Dec	11	26	350

Looking at just the year-end average (~17 incidents/month) would hide the fact that incidents nearly doubled between January and July before declining — a pattern that demands investigation.

💡 **The Insight:** Time series data lets us see the narrative of public safety, not just the final score.

Key Takeaway: Always ask not just "how many" but "when" — the timing tells the real story.